

Evaluation of an automatic ocular filtering method for awake spontaneous EEG signals based on Independent Component Analysis

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Abstract—Electroencephalographic artifacts associated with eye movements are a potential source of error in the EEG analysis when its interpretation is performed for evaluating the influence of drugs and the diagnosis of neurological disorders. In this study, a new automatic method for artifact filtering based on Independent Component Analysis (ICA) is proposed. Automatic artifact identification is based on frequency domain and scalp topography aspects of the independent components. A comparative study between ICA and the ‘gold standard’ method based on linear regression analysis is performed. The latter does not take into account the mutual contamination between EEG and electrooculographic activity, reducing not only the ocular movements but also interesting cerebral activity, mainly in anteriorly placed electrodes. This limitation is overcome by ICA and the efficiency of this approach is shown for a double-blind, placebo-controlled crossover drug trial in healthy volunteers.

Keywords—EEG, ICA, ocular artifacts, regression analysis

I. INTRODUCTION

Quantitative analysis and interpretation of the human electroencephalogram (EEG) has been found to be a valuable method in clinical neuropsychopharmacology as a classification instrument whereby one may objectively evaluate a drug bioavailability at the human brain [1]. A widely used method for evaluating the influence of drugs and in the diagnosis of psychiatric and neurological disorders is based on the changes obtained in some spectral EEG variables. There are four spectral bands of clinical interest in pharmaco-EEG studies: delta (1.3-3.5 Hz), theta (3.5-7.5 Hz), alpha (7.5-13 Hz) and beta (13-30 Hz). Not everyone agrees on the exact boundaries of these rhythms and many people subdivide more these bands.

Benzodiazepines are currently the pharmacological treatment most commonly used for anxiety disorders. It is well-known the effects that benzodiazepines produce on these EEG bands (increase of delta and beta, decrease of alpha power defined between 7.5 and 10.5 Hz) [2].

It is known that non-cortical activity contributes to the EEG recordings. Electrooculogram (EOG), muscle activity and electrocardiogram are the most important non-cortical sources [3]. A critical point with relevant importance in EEG data processing is the careful treatment of these artifacts that contaminate the EEG signals and that could lead up to wrong results and conclusions. The most relevant case is the ocular artifact. It generates a large-amplitude

alternate current field, which can be recorded by electrodes near the eyes when the globe rotates about its axis. In general, artifact minimization is preferable to artifact rejection, since no loss of data is entailed when limited data are available, or eye movements occur too frequently [4].

The approach most commonly used to minimize eye movement artifacts is based on regression analysis in time or frequency domain, subtracting from the EEG signal a part of the contribution of the EOG due to the propagation of the electrical eye activity towards the EEG data. Regression methods depend on having a good regression channel, and they share an inherent weakness which is that spread of excitation from eye movements and EEG signals is bidirectional. Therefore, whenever regression-based artifact removal is performed, relevant cerebral information contained in the EOG channels is also cancelled out in the corrected EEG [5].

Other approaches are based on a linear decomposition of the EEG and EOG signals into source components, in order to solve the limitations of regression methods. These other approaches identify artifactual components, and then reconstruct the EEG signals without them. One of these methods is the Independent Component Analysis (ICA), which was developed in the context of blind source separation problems in order to obtain source components that are approximately independent. ICA provides a method for artifact removal where it is neither necessary an accurate model of the process that generates the artifacts nor the specified observation intervals that contain mainly the artifact [6].

The aim of the present study is to describe an automatic artifact filtering method based on ICA with special emphasis on its advantages in relation to the regression method in pharmaco-EEG studies.

II. METHODOLOGY

A. Signals, medication and design

Twelve healthy volunteers, six males and six females, aged between 20 and 32 years (mean age 23.75 years) were included in the trial. The study was a double-blind, placebo-controlled, dose-increasing, cross-over experiment. Single oral doses of 0.25, 0.5 and 1 mg alprazolam or placebo were administered in a balanced order. Experimental days were

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separated by 1-week washout period. The volunteers were allowed to take any psychoactive drugs neither two weeks before nor during the period of the study.

A 3 minute vigilance-controlled EEG with eyes closed was recorded before (PRE) as well as 30min, 45min, 1h, 1h30min and 2h30min after drug intake (8 a.m.). During vigilance-controlled spontaneous EEG recordings, the experimenter tried to keep the volunteers alert; they were aroused by acoustic stimulation as soon as drowsiness patterns appeared in the EEG. In addition to vertical and horizontal EOG, 19 EEG channels (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1 and O2 referenced to averaged mastoid electrodes, according to the international 10-20 system) were recorded with a sampled rate of 500 Hz.

B. Filtering Methods

B.1. Regression analysis

Two step automatic artifact filtering approach used to reduce ocular artifacts based on regression is described in [7]. In the first step, EOG reduction using grand mean regression factors and artifact rejection using weighted thresholds are yielded in order to validate individual regression factors (α for vertical EOG, VEOG, and β for horizontal EOG, HEOG). It is assumed that the recorded EEG signals (EEG_{raw}) are a linear, time-invariant superposition of different sources:

$$EEG_{corr}(t) = EEG_{raw}(t) - \alpha \cdot VEOG(t) - \beta \cdot HEOG(t) \quad (1)$$

In the second step, EOG subtraction using these individual regression factors and artifact rejection using restrictive thresholds are applied in order to select epochs free of artifacts.

B.2. Independent Component Analysis

ICA is a statistical signal processing technique whose goal is to express a set of random variables as a linear combinations of statistically independent component variables. A way of formulating the ICA problem is to consider the parametric estimation of the following generative model for the data:

$$x = As \quad (2)$$

where x is a matrix composed by n row vectors, s is a matrix composed by m row vectors assumed mutually independent, and A is a $n \times m$ matrix which must be estimated. In other words, there are some mixtures x_i of some original sources signals s_j called independent components (IC). Nothing is known about the sources s_j or the mixing process A . The objective is to estimate the original sources [6].

Several different estimation principles and algorithms for ICA have been recently introduced. Taking into account

that the sources are time signals, the algorithm selected in this study is a method which uses a time structure called Algorithm for Multiple Unknown Signals Extraction (AMUSE) [8]. In fact, AMUSE is an algorithm that applies second-order statistics of the IC. It was developed to circumvent shortcomings that present some fourth-order blind identification algorithms when the sources are gaussian.

In (2), rows of the observed matrix x are the raw EEG and EOG signals recorded at different electrodes. Rows of the output data matrix $u = Wx$ are the estimated source signals (time courses of activations of the IC), and the columns of the inverse matrix W^{-1} provide the projection strengths of the respective component onto the scalp sensors. The scalp topographies of the IC allow to examine the evidence for their biological origin and the determination if an IC is related or not with an artifact. After detecting the IC related to EOG interference, corrected EEG can be reconstructed from the remaining components by zeroing out the corresponding rows of the matrix u .

Automatic artifact identification is based on frequency and scalp topography aspects of the IC. The criteria to remove an IC related to an eye movement artifact are empirically defined as:

- i) Relative power in the delta frequency band of this IC must exceed a fixed high percentage.
- ii) Projection strength (normalized to the maximum) of this IC onto the EOG electrodes must be higher than a threshold.
- iii) The projection strengths of the IC onto the EEG electrodes have to follow a gradient depending on the artifact (if it corresponds to a vertical movement: $Fp > F > C > P > O$; or to a horizontal movement: $F > C > P > O$).
- iv) Maximum of the projection strengths onto the EEG electrodes must be higher than a threshold, which depends if the artifact corresponds to a vertical or horizontal eye movement. This condition is applied in order to validate the same automatic algorithm with the sleep EEG signals in the future, and not to consider K-complexes and delta waves as EOG artifacts.

C. Spectral and statistical analysis

Spectral analysis is performed in corrected 5s epochs. Power spectral density (PSD) function is calculated by means of periodogram using a Hanning window to prevent leakage of spectral power. PSD functions for all artifact-free EEG epochs are averaged in every observed times of the experiment. Several variables are calculated from the mean PSD functions: total power, absolute and relative power in 8 different frequency bands (delta [1.3-3.5 Hz], theta [3.5-7.5 Hz], alpha1 [7.5-10.5 Hz], alpha2 [10.5-13 Hz], beta1 [13-16 Hz], beta2 [16-20 Hz], beta3 [20-25 Hz] and beta4 [25-30 Hz]).

Statistical analysis of EEG recordings is performed following the IPEG (International Pharmac-EEG group) guideline on statistical design and analysis of pharmacodynamic trials [9]. In short, descriptive tests, preferably of simple null hypotheses such as equality of two treatment effects, are performed at all observation times, locations and measurements (variables). Topographic brain maps of drug-induced changes compared to placebo-induced alterations based on t-tests are also carried out in order to show drug effects in each of the 17 EEG variables.

III. RESULTS

ICA decomposition is performed on 5-second EEG epochs, using the AMUSE algorithm from the ICALAB toolbox v2.1. for Matlab [10]. As an example, Fig. 1 shows EEG and EOG signals corresponding to 5-second epoch before and after filtering processes (regression analysis and ICA) to reduce ocular activity. Because the propagation factors are very high at the anterior electrodes, regression-based artifact removal approach also removes neural activity in common to EEG and EOG data, as it can be seen in the corrected Fp1 and F7 signals.

The influence of ICA minimization of ocular artifacts compared to regression method is evaluated by means of PSD functions at different observation times in recordings with placebo and drug doses. Fig. 2 shows, as an example, PSD functions obtained from electrodes Fp1, F7 and O1 at 1h after alprazolam administration (expected time for maximum effect), before and after applying regression and ICA filtering methods. Regression-based artifact removal not only reduces ocular frequency components but also removes interesting cerebral activity, mainly in anteriorly placed electrodes.

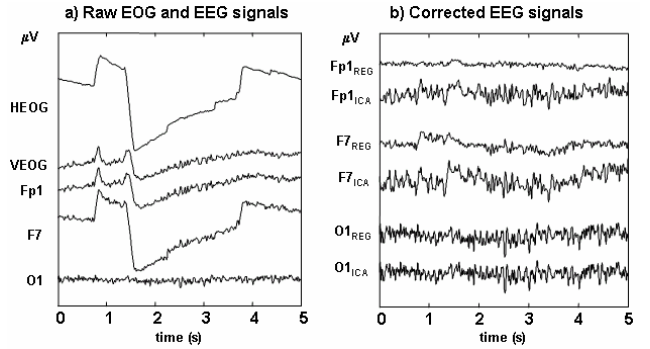


Fig. 1. a) A five second epoch of raw EEG and EOG data. b) EOG Artifact-free EEG signals.

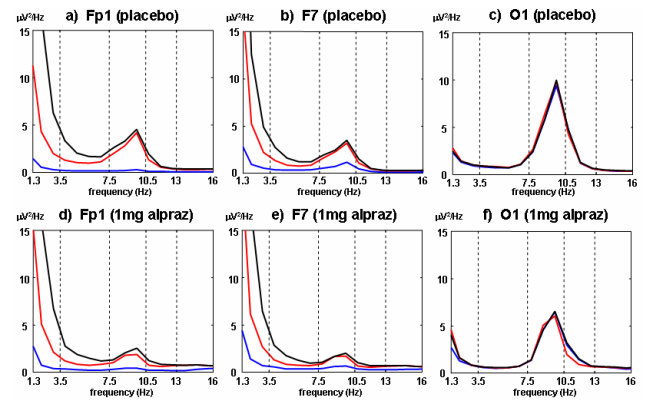


Fig. 2. Average PSD functions at the maximum effect of alprazolam (1h after drug intake): raw EEG signals (black), after ICA (red), and after regression filtering (blue). Vertical dotted lines separate frequency bands.

Topographic brain maps of alprazolam-induced changes compared with placebo-induced alterations based on t-tests are analyzed in order to show the impact on drug effects in

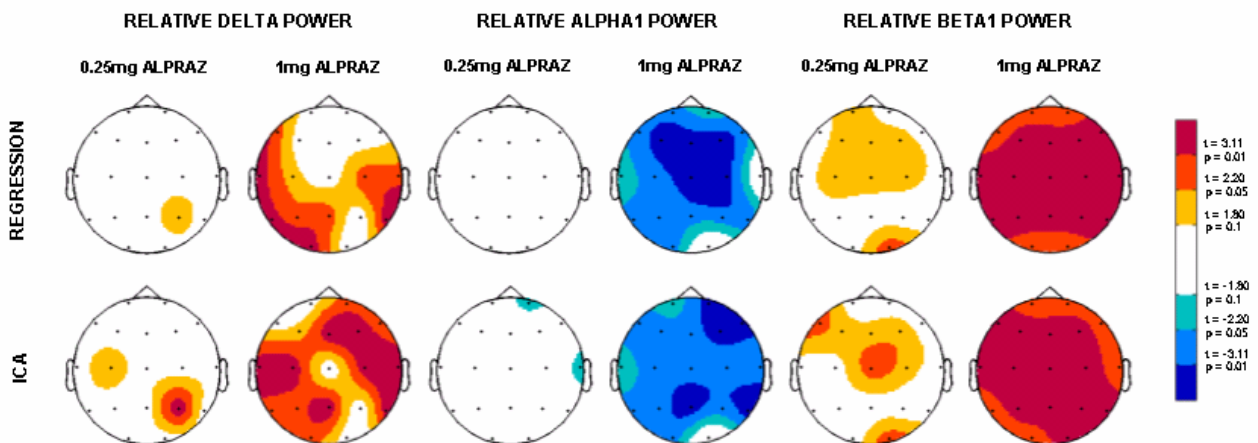


Fig 3. Maps of topographic pharmaco-EEG changes in the relative power of the delta, alpha1 and beta1 activities after alprazolam 0.25 and 1 mg compared to placebo at 1 hour after administration vs baseline values for regression filtering (upper) and ICA method (lower). The seven-colour scale represents drug-induced changes based on t-values expressed on p-values. Electrode positions are indicate by black dots.

some of the 17 EEG variables. Maps of topographic pharmaco-EEG changes at 1 hour after drug intake are presented as an example for the two filtering methods in Fig. 3. Relative delta power increases in more electrodes with ICA filtering and similar maps are obtained between both methods in the other frequency bands.

Statistical differences for some absolute variables are shown in Fig. 4 as an example in order to compare regression and ICA-based artifact methods. ICA-based method preserves more cerebral activity in the anteriorly placed electrodes than regression analysis.

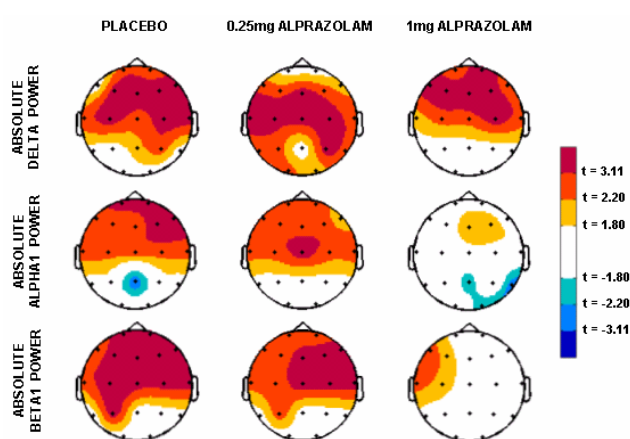


Fig. 4. Significance probability maps showing differences between regression analysis and ICA in the absolute delta, alpha1 and beta1 powers at 1h after placebo, 0.25 and 1mg alprazolam intake. For technical description of the maps and explanation of the colour key see Fig. 3.

IV. DISCUSSION

Ocular potentials are clearly a nuisance in studies of the influence of drugs by means of the quantitative analysis of the human EEG, due to PSD functions of the EEG and EOG signals overlap in frequency.

In this study, an empirically automatic ICA-based artifact reduction method is applied for evaluating the bidirectional contamination between the EEG and EOG signals. This mutual contamination between EEG and EOG data is not taking into account by the ‘gold standard’ artifact removal method based on regression analysis used in pharmaco-EEG trials.

Time and frequency results indicate that ICA preserves and recovers more brain activity than regression in decomposing real EEG data. Otherwise, similar results are obtained for both artifact removal methods in the topographic maps of alprazolam-induced changes compared with placebo-induced alterations, except for relative delta power band with an increment in more electrodes with ICA than regression method.

This low level on drug results impact could be due to the type of design used in pharmaco-EEG studies with

healthy volunteers, that is within-subjects and within-time comparisons. However, different results could be obtained, for example, if differences in EOG activity could have part of drug profile effects.

Significance probability maps of the differences between both methods reveal more cerebral activity mainly in the frontal placed electrodes in the corrected EEG signals by means of ICA. This result is coherent with the distribution across the scalp of the electrical potentials generated by the eye activity. Moreover, the anterior EEG activity is subtracted along with the EOG activity when calculating corrected EEG signal by means of regression method.

ICA appears to be a generally applicable and effective method for removing ocular artifacts from spontaneous EEG recordings since it simultaneously separates both the EEG activity and its artifacts into IC based on the statistics of the recorded signals, without relying on the availability of clean reference channels. This avoids the problem of mutual contamination between artifacts and removing useful information.

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